Chapter 6, Problem 50P

Problem

An energy-storage network consists of series-connected 16- and 14-mH inductors in parallel with series-connected 24- and 36-mH inductors. Calculate the equivalent inductance.

Step-by-step solution

Step 1 of 2

Consider that the series connection of 16 mH and 14 mH inductors is in parallel connection with the series connection of 24 mH and 36 mH inductors.

Draw the energy-storage network with inductors.

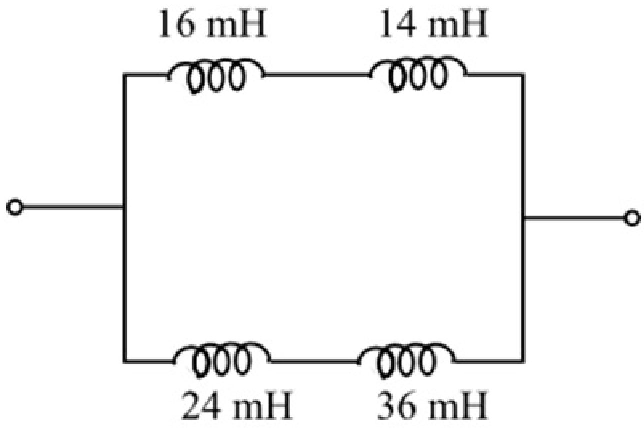

Figure 1

Step 2 of 2

Determine the equivalent inductance of the series combination of inductances.

$$\begin{aligned} L_{eq1} &= 16\text{ mH} + 14\text{ mH} \\ &= 30\text{ mH} \end{aligned}$$

$$\begin{aligned} L_{eq2} &= 24\text{ mH} + 36\text{ mH} \\ &= 60\text{ mH} \end{aligned}$$

Determine the equivalent inductance of the parallel combination of series connected inductances, L_{eq1} and L_{eq2}.

$$\begin{aligned} L_{eq} &= \frac{L_{eq1}L_{eq2}}{L_{eq1} + L_{eq2}} \\ &= \frac{(30\text{ mH})(60\text{ mH})}{30\text{ mH} + 60\text{ mH}} \\ &= \frac{(30\text{ mH})(60\text{ mH})}{90\text{ mH}} \\ &= 20\text{ mH} \end{aligned}$$

Thus, the equivalent inductance of the network is $\boxed{20\text{ mH}}$.